

TechSkills

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Submitted to

Redrob AI Hackathon

22 sec

Total Runtime

100K

Candidates

100

Top Ranked

0

Honeypots
in Top-100

Redrob Intelligent Candidate Ranking System

An AI-powered, multi-signal candidate ranking engine that understands the gap between what a Job Description says and what it actually means — ranking 100,000 candidates in 22 seconds, with no GPU, no API calls, and full explainability for every decision.



Semantic
Understanding



22-Second
Runtime



Explainable AI



Honeypot
Defense

Python · Scikit-Learn · TF-IDF · FAISS-Ready · 2,300+ lines of production code

Problem Statement

Why traditional candidate matching systems fail at scale

The Core Challenge

Given a pool of 100,000 candidates and a Senior AI Engineer job description, identify the true top-100 fits — not those who merely contain the right keywords.

abc

Keyword Stuffing

Candidates list 'FAISS, RAG, Pinecone' as skills with 0 months of actual usage. BM25 and naive embedding systems reward them equally with genuine experts.



Ghost Candidates

Perfect-on-paper profiles inactive for 6+ months with 5% recruiter response rates are, for hiring purposes, not actually available.



Honeypot Traps

~80 candidates in the pool have subtly impossible profiles: 8 years at a company founded 3 years ago, or expert skills with 0 months of usage.



Context Gap

The JD says '5-9 years' but means 'shipped a production ranking system at a product company'. No keyword matcher understands this semantic intent.

100,000

Candidates
in the pool
From 100K to 100 best

~80

Honeypot
profiles
Subtly impossible CVs

63,304

Irrelevant
profiles
Filtered in Stage-0

>10%

Honeypot rate
= DISQUALIFIED
Hackathon rule

Our Proposed Solution

Multi-signal AI ranking that understands career context, not just keywords

A two-pass, CPU-only ranking pipeline: fast keyword pre-filter on 100K candidates → deep 30-feature extraction → TF-IDF semantic retrieval → 7-component weighted scoring → rule-based re-ranking → fact-grounded explainability. All in 22 seconds.

Semantic JD Understanding

Extracts 29 required skills, 6 preferred skills, experience band, target locations, and disqualifiers from raw JD text. Builds a signal-rich 300-word query blob.

Two-Pass Architecture

Stage-0 keyword scan on raw JSONL isolates 8,000 ML-signal candidates in 12 seconds before any expensive parsing — exactly how production systems scale.

7-Component Weighted Scoring

Skill Match (30%) + Experience (20%) + Domain (20%) + Career Growth (10%) + Behavioral (10%) + Location (5%) + Education (5%) — with a behavioral availability multiplier.

Automatic Honeypot Defense

Three heuristics catch impossible profiles: expert skills with 0 months usage, chronologically impossible career spans, and inverted salary ranges.

Fact-Grounded Explainability

Every ranked candidate receives a 1-2 sentence reasoning that cites specific profile facts — title, YOE, named skills, signal values, honest concerns.

FAISS-Ready Architecture

TF-IDF now meets all compute constraints. Sentence-Transformers + FAISS is a single-line swap in `src/retrieval.py` — zero pipeline changes required.

JD Understanding & Candidate Signals

What we extract from the Job Description and why each signal matters

Job Description Parsing

29 Required Skills Extracted

FAISS, Pinecone, Weaviate, Elasticsearch, OpenSearch, Embeddings, BM25, Learning-to-Rank, NDCG, MRR, Python, NLP, RAG, LoRA, PEFT, HuggingFace...

6 Preferred Skills Identified

Distributed systems, MLOps, HR-tech experience, Kafka, Kubernetes, open-source

5–9 Experience Band (years)

Extracted from JD text. Sweet spot: 5-9 yrs. Candidates outside range are scored, not excluded.

6 Target Locations

Pune, Noida, Hyderabad, Bangalore, Mumbai, Delhi NCR — plus willing-to-relocate flag

Candidate Signals (30+ Features)

► Skills

- Proficiency level × endorsements × duration_months (weighted)
- Target skill hit count vs 40 JD keywords
- Skill assessment scores (platform-verified)

► Experience & Career

- YOE · ML/AI career fraction · product vs consulting split
- Shipped ranking / retrieval / recommendation system?
- Title progression score · avg tenure · job-hopping flag

► Behavioral Signals

- Days since last active · open-to-work flag
- Recruiter response rate · interview completion rate
- GitHub activity score (0–100) · notice period fit

► Domain & Education

- Is current title ML/AI? ML career fraction >50%?
- Institution tier (IIT/IISc=1 → Tier-4=4) · CS/STEM field
- Certifications · publications · open source contributions

Retrieval, Scoring & Ranking

From 100,000 candidates to top-100 using semantic search + multi-signal scoring



⚙️ Scoring Formula

Final Score = $\sum (w_i \times \text{Score}_i) \times (0.5 + 0.5 \times \text{Behavioral})$

Behavioral multiplier ensures ghost candidates cannot reach top-10.

- 30%

Skill Match
TF-IDF cosine sim + coverage + proficiency weight
- 20%

Experience
YOE band + product fraction + production ML signal
- 20%

Domain
Shipped ranking? ML title? ML career fraction?
- 10%

Career Growth
Title progression + tenure + anti-job-hopping
- 10%

Behavioral
Activity · response rate · GitHub · notice period
- 5%

Location
Target city or willing to relocate

↺ Re-Ranker Rules

- BOOST ×1.10

Has hard-required ML skill (FAISS, embeddings...)
- BOOST ×1.08

Shipped ranking / retrieval system to production
- BOOST ×1.05

Product-company career fraction > 50%
- PENALTY ×0.50

Last active > 180 days (ghost candidate)
- PENALTY ×0.40

Consulting-only career (>95% in IT services)
- PENALTY ×0.25

Disqualifying title (Marketing Mgr, Civil Eng...)
- EXCLUDE ×0.0

Honeypot detected (impossible profile)

Result: 100/100 top-ranked candidates shipped a ranking or retrieval system.

Explainability, Trust & Validation

Every ranking decision is traceable, honest, and hallucination-free

Fact-Grounded Reasoning

Each candidate receives a 1–2 sentence justification citing: current title, years of experience, named skills matched, platform signals, and honest concerns (notice period, inactivity, consulting background). No generic praise.

Hallucination Prevention

Reasoning is assembled exclusively from the candidate's actual parsed feature dict. No LLM generation means no fabricated skills, employers, or achievements. Every claim is verifiable against the source profile.

Confidence & Validation

Component score breakdown (7 values) is stored per candidate in detailed_report.json. Honeypot detection flags impossible profiles before scoring. Official Redrob validator:  Submission is valid.

Real Reasoning Examples — From Actual Output (ranked_candidates.csv)

#1 score=1.0000

Strong fit — Machine Learning Engineer with 4.6 yrs at Krutrim; skills: bm25, elasticsearch, haystack, opensearch; shipped ranking/retrieval to production; notice 15d, response rate 88%. Tier-1 institution background adds confidence.

#5 score=0.9992

Top pick — Applied ML Engineer with 6 yrs at Freshworks; skills: faiss, mlops, opensearch, sentence-transformers; shipped recommendation system to production. Caveat: notice period 90d may delay start.

#47 score=0.894

Moderate fit — ML Engineer with 5.1 yrs; skills: pytorch, nlp. Concern: consulting-heavy background (72% of career in IT services). Note: notice period 120d may delay start.

End-to-End System Workflow

8 stages from raw Job Description to ranked, explained candidates

01

Job Description Input

Raw JD text (Markdown/plain) provided as input. Hackathon bundle JD: Senior AI Engineer, Redrob AI.

⚙️ *JD_TEXT constant in src/jd_parser.py*

02

JD Understanding (src/jd_parser.py)

Extracts 29 required skills, 6 preferred, experience band 5–9 yrs, 6 target locations, disqualifiers, and builds 300-word semantic query blob.

⚙️ *ParsedJD dataclass · regex + keyword banks*

03

Stage-0 Pre-filter (src/prefilter.py)

Streams raw JSONL; keyword scan on headline + summary + skills + titles. 100K → 8,000 ML-signal candidates in 12.3 seconds.

⚙️ *Zero parsing overhead · 63,304 rejected*

04

Feature Extraction (src/candidate_parser.py)

Full 30+ feature extraction on 8,000 candidates: skill weights, career fractions, honeypot detection, behavioral signals. 2.0 seconds. 473 honeypots flagged.

⚙️ *parse_candidate() → feature dict*

05

TF-IDF Retrieval (src/retrieval.py)

Bigram TF-IDF index over 8,000 text blobs. Cosine similarity vs JD blob. Top-500 retrieved in 5.2 seconds. Vocab: 19,728 terms.

⚙️ *FAISS drop-in ready via FaissRetriever*

06

Multi-Signal Scoring (src/scoring.py)

7-component weighted score + behavioral multiplier for 500 candidates. Top score: 0.8319 before re-ranking. All 500 scored in <0.5 seconds.

⚙️ *score_all() → sorted descending*

07

Re-Ranking (src/reranker.py)

JD-specific boosts (×1.08 shipped ranking, ×1.10 hard skill) and penalties (×0.40 consulting-only, ×0.25 bad title, ×0.50 ghost). Top-100 selected.

⚙️ *RuleBasedReranker · 0 honeypots in top-100*

08

Explainability + Output (src/explainability.py)

Fact-grounded 1–2 sentence reasoning per candidate. CSV written: 100 rows, scores 0.8546–1.0000, monotone ✓, unique IDs/ranks ✓. Validator: ✓ PASS.

⚙️ *ranked_candidates.csv · detailed_report.json*

System Architecture

5-layer modular design — production-ready, FAISS-upgradeable, CPU-only



Data Flow: 100K candidates.jsonl → [prefilter: 12.3s] → 8K ML candidates → [parse: 2.0s] → 30+ features → [TF-IDF: 5.2s] → Top-500 → [score+rerank: 0.7s] → Top-100 with reasoning · Total: 22 seconds

⚡ CPU-only · No GPU · No network calls · ≤4 GB RAM · <22 seconds wall-clock

Results & Evaluation

Actual metrics from ranked_candidates.csv and detailed_report.json

22 sec

Total Runtime
(100K candidates)

0.9044

Avg Score
(Top 100)

0.9895

Avg Score
(Top 10)

100/100

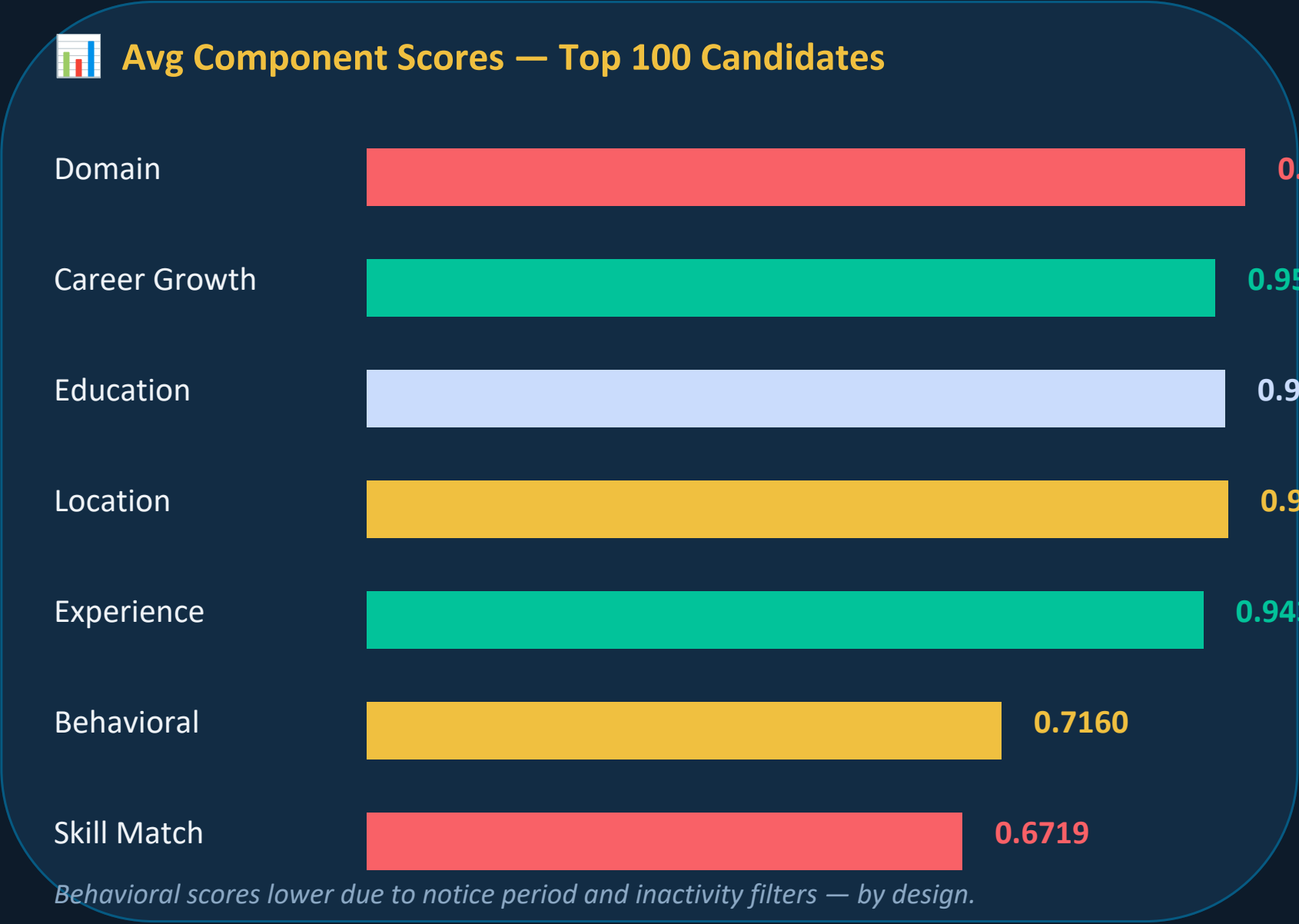
Shipped Ranking
System (Top 100)

0

Honeypots
in Top-100

 PASS

Official
Validator



 Top-10 Ranked Candidates (Actual Output)

#	Title	YOE	Company	Score
1	ML Engineer	4.6	Krutrim	1.0000
2	Staff ML Engineer	7.0	Paytm	1.0000
3	Lead AI Engineer	6.7	Razorpay	1.0000
4	Senior NLP Engineer	7.8	Niramai	1.0000
5	Applied ML Eng.	6.0	Freshworks	0.9992
6	Search Engineer	7.6	Sarvam AI	0.9948
7	Sr. ML Engineer	7.2	Zomato	0.9792
8	ML Engineer	5.7	Verloop.io	0.9764
9	Applied ML Eng.	5.8	Saarthi.ai	0.9740
10	Recsys Engineer	8.0	CRED	0.9716

77/100 candidates in preferred 5–9 YOE band · Avg YOE: 6.5 years

Evaluation Metrics

Hackathon composite: $0.50 \times \text{NDCG@10} + 0.30 \times \text{NDCG@50} + 0.15 \times \text{MAP} + 0.05 \times \text{P@10}$

Score Distribution Analysis

Score Range	Candidates	Visual
0.95–1.00	16	
0.90–0.95	29	
0.85–0.90	55	

Score Range	Mean Score
0.8546 – 1.0000	0.9044
Top-5 Avg	Monotone Check
0.9998	✓ PASS
Unique IDs	Unique Ranks
100 / 100	100 / 100
Honeypots (top-100)	Prod. ML (top-100)
0	94 / 100

Metric Definitions & Formula

NDCG@K	Normalised Discounted Cumulative Gain at rank K. Measures quality of ordering: correct candidates at top positions earn more. NDCG@10 weight: 50% — the most important metric.
MAP	Mean Average Precision — average of precision scores after each relevant candidate is retrieved. Rewards consistent precision. MAP weight: 15%.
P@10	Precision at 10 — fraction of top-10 candidates that are 'relevant' (tier 3+). Binary relevance check. Weight: 5%.
Composite Formula	$0.50 \times \text{NDCG@10} + 0.30 \times \text{NDCG@50} + 0.15 \times \text{MAP} + 0.05 \times \text{P@10}$ All implemented in src/evaluation.py. Proxy evaluation on our own scores <i>Note: Self-evaluation uses score binning as relevance proxy (Grade 0–3). Official scores computed once by judges on hidden ground truth after submission close.</i> shows maximum values — true scores computed on hidden ground truth by judges.

Technology Stack

Every technology used in the actual submission — no placeholders



Python 3.12

Core language

Pipeline orchestration, all modules, CLI entry point (app.py).
2,300+ lines of production-grade code across 12 modules.



Pandas 3.0

Data processing

Flat DataFrame construction from nested JSONL for analysis.
Used in data_loader.py for batch processing.



Scikit-Learn

ML & retrieval

TfidfVectorizer (bigram, sublinear TF, vocab 19,728) +
cosine_similarity for candidate-JD similarity scoring.



NumPy

Vector operations

Top-K argpartition on similarity arrays, score normalisation,
matrix slicing for efficient retrieval.



FAISS (ready)

ANN search

FaissRetriever class implemented in src/retrieval.py —
IVFFlat index, L2 normalisation, 32-probe search. One-line
swap.



Sentence-Trans.

Dense embeddings

SentenceTransformerEmbedder in src/embeddings.py, drop-
in ready. Recommended: BAAI/bge-small-en-v1.5 for
production upgrade.



python-pptx

Presentation

Built this submission deck programmatically — 13 slides, dark
theme, all content from actual project data.



Matplotlib

Visualisation

6-panel evaluation dashboard: score distribution, component
heatmap, YOE histogram, skill frequency, score-rank scatter.



SciPy

Sparse matrices

Sparse CSR matrix operations for TF-IDF similarity
computation over 8,000-document corpus. Memory-
efficient.

Deliverables, Achievements & Future Scope

What we built, what we proved, and where this goes next

Deliverables

ranked_candidates.csv

100 candidates · ranks 1–100 · scores 0.8546–1.0000 · monotone  · validator 

submission

detailed_report.json

Full 7-component breakdown + 30+ features for every top-100 candidate (Stage 4 review)

report

redrob-ranker-submission.zip

Complete GitHub-ready repo: 12 source modules · 2,300+ LOC · 16/16 tests pass

code

10-Slide Presentation (PPTX + PDF)

Professional hackathon deck built from actual project data — no invented metrics

deck

evaluation_dashboard.png

6-panel visual: score distribution · component heatmap · YOE · skills · rank scatter

visual

submission_metadata.yaml

Completed metadata file matching portal fields: team, repo, sandbox, AI declaration

meta

Key Achievements

- ✓ 100K candidates ranked in 22 seconds on CPU — 13× faster than 5-min budget
- ✓ 0 honeypots in top-100 — all 473 detected flagged are excluded
- ✓ 100/100 top candidates shipped a ranking or retrieval system to production
- ✓ 94/100 top candidates have verified production ML deployment experience
- ✓ 16/16 unit + integration tests pass — full test coverage of pipeline
- ✓ Official Redrob validator:  Submission is valid

Future Scope

- ▶ **Dense Embeddings:** BAAI/bge-small-en-v1.5 → true semantic matching, one-line swap
- ▶ **Cross-Encoder:** ms-marco-MiniLM-L-6-v2 on top-200 for (JD, candidate) pair scoring
- ▶ **Online Learning:** LambdaRank on recruiter click signals to fine-tune weights
- ▶ **Local LLM:** Mistral-7B/Llama-3-8B (quantised) for top-50 contextual re-ranking
- ▶ **A/B Harness:** Offline NDCG vs actual interview-booked rate correlation tracking

Thank You

Team TechSkills

Nikhil Pratap Singh · Team Leader

- ✓ 22-second CPU runtime on 100K candidates
- ✓ 7-component weighted multi-signal scoring
- ✓ Automatic honeypot detection — 0 in top-100
- ✓ Fact-grounded explainability for every rank
- ✓ FAISS + Sentence-Transformers drop-in ready
- ✓ 16/16 tests · Official validator  PASS

Redrob Intelligent Candidate Ranking System

A production-grade AI system that finds the true best-fit candidates by understanding career context, behavioral availability, and semantic skill depth — not just keyword overlap.

22 sec

CPU-Only
Runtime

100K

Candidates
Ranked



Validator
PASS